

Theorem 2. The algebraic boundary of the period domain

$\Omega(a, N-a) = \pi/\sin(\pi a/N)$, the sine product, and the period witness W_N

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Study	CYCLORING
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Abstract

The monograph introduces the period functional of the level $\Omega(a,b) = \Gamma(a/N)\Gamma(b/N)/\Gamma((a+b)/N)$ and proves that its boundary is algebraic: $\Omega(a, N-a) = \pi/\sin(\pi a/N)$ — an algebraic multiple of π lying in $\pi Q(\zeta_N)$. The interior periods are genuine Gamma-transcendentals, and the program never uses them algebraically.

The algebraic backbone is the sine product of $2 \sin(\pi m/N) = N$, which yields the closed form of the product of all boundary periods $\pi^{N-1} 2^{N-1}/N$. The diagonal normalized sums $P(a,a)$ generate the period witness W_N — the only periodic number of the level entering the defect chain.

Certificates: stage W1 — the reflection identity with the residual $5.1e-49$ (mpmath, 50 digits) at all levels 7, 9, 15, 30; stage W2 — the sine product with the same residual; the float versions hold $1e-12$.

At a glance

Functional	$\Omega(a,b) = \Gamma(a/N)\Gamma(b/N)/\Gamma((a+b)/N)$
Normalization	$P(a,b) = \Omega(a,b)/\Omega(1,1); P(1,1) = 1$
Boundary	$\Omega(a, N-a) = \pi/\sin(\pi a/N)$ in $\pi Q(\zeta_N)$
Sine product	of $2 \sin(\pi m/N) = N$
Boundary product	$\pi^{N-1} 2^{N-1}/N$
Period witness	$W_N = \sum_{a=1}^{N-1} P(a,a)$

1. The period functional of the level

For integers $a, b \geq 1$ set $\Omega_{a,b} = (\Gamma(a/N)\Gamma(b/N))/(\Gamma((a+b)/N))$ and $P(a,b) = \Omega_{a,b}/\Omega_{1,1}$. The functional is symmetric, positive, and the normalization is canonical: $P(1,1) = 1$ exactly. The period witness of the level is the diagonal sum $W_N = \sum_{a=1}^{N-1} P(a,a)$.

$$\Omega_{a,b} = \frac{\Gamma(a/N) \Gamma(b/N)}{\Gamma((a+b)/N)}, \quad W_N = \sum_{a=1}^{N-1} \frac{\Omega_{a,a}}{\Omega_{1,1}}$$

2. The algebraicity of the boundary

The reflection identity $\Gamma(z)\Gamma(1-z) = \pi/\sin(\pi z)$ at $z = a/N$ gives $\Omega_{a, N-a} = \Gamma(a/N)\Gamma(1-a/N) = \pi/\sin(\pi a/N)$, since $\Gamma(1) = 1$. The number $\sin(\pi a/N)$ is algebraic — it is expressed through ζ_N ; hence the boundary period is an algebraic multiple of π . The interior pairs $a + b \neq N$ admit no such reduction: they are treated as Gamma-transcendentals.

$$\Omega_{a, N-a} = \frac{\pi}{\sin(\pi a/N)} \in \pi \cdot \mathbb{Q}(\zeta_N)^\times$$

3. The sine product

The polynomial $(z^N - 1)/(z - 1) = \prod_{m=1}^{N-1} (z - \zeta_N^m)$ at $z = 1$ gives $\prod_m (1 - \zeta_N^m) = N$; passing to the moduli and the identity $|1 - e^{i\varphi}| = 2\sin(\varphi/2)$ complete the argument: $\prod_m 2\sin(\pi m/N) = N$. Corollary: the product of all boundary periods equals $\pi^{N-1} 2^{N-1}/N$ — a closed arithmetic register of the level.

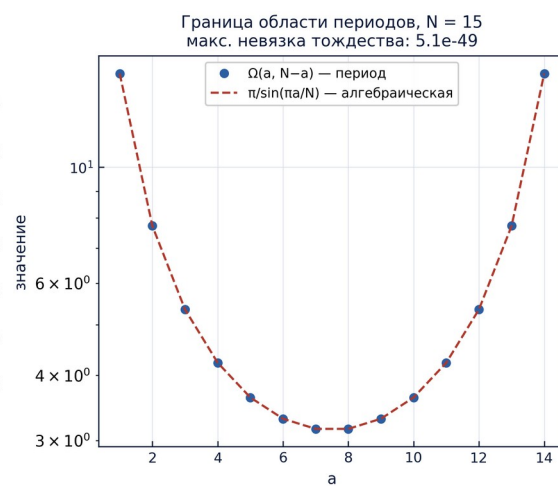
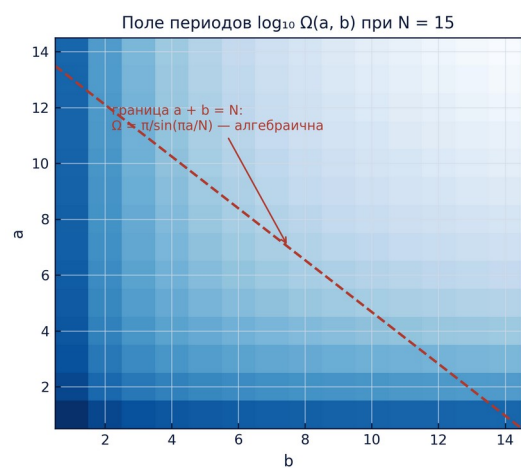
$$\prod_{m=1}^{N-1} 2\sin \frac{\pi m}{N} = N, \quad \prod_{a=1}^{N-1} \Omega_{a, N-a} = \frac{\pi^{N-1} 2^{N-1}}{N}$$

4. Certificates and exact values

Stages W1–W2 hold the mpmath residuals $5.1 \cdot 10^{-49}$ at the levels 7, 9, 15, 30 (all boundary pairs). Exact values at small levels: $\Omega_{1,2} = 2\pi/\sqrt{3}$ at $N = 3$; $\Omega_{1,3} = \pi\sqrt{2}$ and $\Omega_{2,2} = \pi$ at $N = 4$; $\Omega_{3,3} = \pi$ at $N = 6$. The witnesses of the four registered levels are tabulated below.

The period witness W_N (protocol W7, default preset)

N	W_N	N	W_N
7	2.175444	15	2.915447
9	2.416543	30	3.603649



The period field of the level and the algebraic boundary (protocol W2).